

Extremal theory of codegree problems

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Abstract

For an r -graph H , let $\mathcal{C}(H) = \min_S d(S)$, where the minimum is taken over all $(r-1)$ -sets of vertices of H , and $d(S)$ is the number of vertices v such that $S \cup \{v\}$ is an edge of H . Given a family $\mathcal{F}$ of r -graphs, the co-degree Turán number $\text{co-ex}(n, \mathcal{F})$ is the maximum of $\mathcal{C}(H)$ among all r -graphs H which contain no member of $\mathcal{F}$ as a subhypergraph. Define the co-degree density of a family $\mathcal{F}$ to be

$$\gamma(\mathcal{F}) = \limsup_{n \rightarrow \infty} \frac{\text{co-ex}(n, \mathcal{F})}{n}.$$

For each $r \geq 2$, let $\Gamma_r = \{\gamma(\mathcal{F}) : \mathcal{F} \text{ is a family of } r\text{-graphs}\}$. The determination of $\gamma(\mathcal{F})$ for every graph family $\mathcal{F}$ is equivalent to the Erdős-Simonovits-Stone theorem, which also implies that $\Gamma_2 = \{0, \frac{1}{2}, \frac{2}{3}, \dots, \frac{k-1}{k}, \dots\}$. Our main result states that for each $r \geq 3$, not only does Γ_r have infinitely many accumulation points, but in fact Γ_r is dense in $[0, 1)$. This phenomenon is similar to the existence of real numbers that are not jumps in hypergraphs, proved by Frankl and Rödl [11].

Several other results about the parameter γ are provided parallel to those of the classical Turán theory. One of these proves the existence, by explicit construction, of finite r -graph families $\mathcal{F}$ satisfying $0 < \gamma(\mathcal{F}) < \min_{F \in \mathcal{F}} \gamma(F)$ for every $r \geq 3$.

1. Introduction

All hypergraphs discussed in this paper are finite and simple. An r -graph H (for $r \geq 2$) is a hypergraph whose edges all have size r . Write $V(H)$ and $E(H)$ for the vertex set and edge set of H , respectively, with $e(H) = |E(H)|$. The notation H_n indicates that $|V(H_n)| = n$.

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Given a family $\mathcal{F}$ of r -graphs, the r -graph G_n is $\mathcal{F}$ -free if it contains no member of $\mathcal{F}$ as a (not necessarily induced) subhypergraph. The *Turán number* $\text{ex}(n, \mathcal{F})$ is the maximum of $e(G_n)$ over all $\mathcal{F}$ -free r -graphs G_n . The limit $\lim_{n \rightarrow \infty} \text{ex}(n, \mathcal{F}) / \binom{n}{r}$ exists by a standard averaging argument of Katona-Nemetz-Simonovits [15], and is often called the *Turán density* and written $\pi(\mathcal{F})$.

Classical extremal graph theory began with Turán's theorem, which determines $\text{ex}(n, F)$ (and therefore $\pi(F)$) when F is a complete graph. The celebrated Erdős-Simonovits-Stone theorem (**ESS**) [8, 10] generalizes Turán's theorem. It states that for any graph F with chromatic number $\chi(F)$, and any $\varepsilon > 0$, there exists $N > 0$ such that every graph G_n with $n > N$ and $e(G) \geq (1 + \varepsilon)(1 - \frac{1}{\chi(F)-1})\binom{n}{2}$ contains a copy of F . It is easy to see that **ESS** implies that $\pi(\mathcal{F}) = \min_{F \in \mathcal{F}} 1 - \frac{1}{\chi(F)-1}$ for any graph family $\mathcal{F}$.

Although the Turán problems study the largest size of graphs not containing certain subgraphs, Turán's theorem and **ESS** can also be viewed as theorems on the largest possible minimum degree of such graphs. For example, letting $\delta(G)$ denote minimum degree in a graph G , **ESS** is equivalent to the following statement:

For any graph F and $\varepsilon > 0$, there exists $N > 0$ such that any graph G_n with $n > N$ and $\delta(G_n) \geq (1 + \varepsilon)(1 - \frac{1}{\chi(F)-1})n$ contains a copy of F .

To see this, we first note that an n -vertex graph with minimum degree cn has at least $c\binom{n}{2}$ edges. On the other hand, for fixed $c, \varepsilon > 0$, every graph G_n (n sufficiently large) with $(c + \varepsilon)\binom{n}{2}$ edges contains a subgraph G' on $m \geq \varepsilon^{1/2}n$ vertices with minimum degree at least cm (see, e.g., [2] p. 121).

What about minimum “degree” extremal problems on hypergraphs? We must first clarify how to define *degree* in hypergraphs. If we consider the usual degree $d(v)$ of a vertex v , defined as the number of edges containing v , then the minimum degree problem is again essentially equivalent to the Turán problem, which is well-studied and known to be extremely hard. Even the reason is the same as before: by averaging arguments, an n -vertex r -graph with minimum degree $c\binom{n-1}{r-1}$ has at least $c\binom{n}{r}$ edges. On the other hand, suppose that $c, \varepsilon > 0$, let G_n be an r -graph with at least $(c + \varepsilon)\binom{n}{r}$ edges. We claim that G_n contains a subgraph G' on $m \geq \varepsilon^{1/r}n$ vertices with minimum degree at least $c\binom{m}{r-1}$. Otherwise, we sequentially remove the vertices $x_1, x_2, \dots, x_{n-\ell}$ whose degree (among the remaining vertices) $d(x_i) < c\binom{n-(i-1)}{r-1}$ for $\ell = \lfloor \varepsilon^{1/r}n \rfloor$. Consequently the number of edges on the remaining ℓ vertices is more than

$$(c + \varepsilon)\binom{n}{r} - \sum_{j=\ell+1}^n c\binom{j}{r-1} = \varepsilon\binom{n}{r} + c\binom{\ell+1}{r} - c\binom{n}{r-1} > \varepsilon\binom{n}{r} > \binom{\ell}{r},$$

which is impossible (the first inequality needs n to be large).

This motivates us to consider another extension of degree in hypergraphs: *co-degree*. Given an r -graph G and a set $S \subset V(G)$ with $|S| = r - 1$, we denote by $N(S)$ or $N_G(S)$ the set of $v \in V(G)$ such that $S \cup \{v\} \in E(G)$. The *co-degree* of S is $d(S) = d_G(S) = |N(S)|$. When $S = \{v_1, \dots, v_{r-1}\}$, we abuse notation by writing $N(v_1, \dots, v_{r-1})$ and $d(v_1, \dots, v_{r-1})$.

Let $\mathcal{C}(G) = \min\{d(S) : S \subset V(G), |S| = r - 1\}$ denote the minimum co-degree in G , and let $c(G) = \mathcal{C}(G)/|V(G)|$.

One purpose of this paper is to show that, for hypergraphs, the co-degree extremal problem is different from the Turán problem. On the other hand, since co-degrees reduce to degrees when the uniformity $r = 2$, our results on co-degree (for all $r \geq 2$) also reveal some similarities in the graph and hypergraph cases.

Definition 1.1. Let $\mathcal{F}$ be a family of r -graphs. The co-degree Turán number $\text{co-ex}(n, \mathcal{F})$ is the maximum of $\mathcal{C}(G_n)$ over all $\mathcal{F}$ -free r -graphs G_n . The co-degree density of $\mathcal{F}$ is

$$\gamma(\mathcal{F}) := \limsup_{n \rightarrow \infty} \frac{\text{co-ex}(n, \mathcal{F})}{n}.$$

Remark. Strictly speaking, one should divide by $n - (r - 1)$ instead of n in the definition of $\gamma(\mathcal{F})$. However, since r is fixed and $n \rightarrow \infty$, this will not change any of our results on $\gamma(\mathcal{F})$, and so we prefer the technically simpler version above.

The argument in [15] shows that $\text{ex}(n, \mathcal{F})/\binom{n}{r}$ is non-increasing in n , and therefore one obtains that $\pi(\mathcal{F}) = \lim_{n \rightarrow \infty} \text{ex}(n, \mathcal{F})/\binom{n}{r}$ exists. Although we could not prove that $\text{co-ex}(n, \mathcal{F})/n$ (or $\text{co-ex}(n, \mathcal{F})/(n - r + 1)$) is non-increasing, we do prove that $\lim_{n \rightarrow \infty} \text{co-ex}(n, \mathcal{F})/n$ exists.

Proposition 1.2. $\gamma(\mathcal{F}) = \lim_{n \rightarrow \infty} \text{co-ex}(n, \mathcal{F})/n$ for all r -graph families $\mathcal{F}$.

Recall that for an r -graph G_n , $\mathcal{C}(G_n) \geq \delta n$ implies that $e(G_n) \geq \delta \binom{n}{r}$. This implies that $\gamma(\mathcal{F}) \leq \pi(\mathcal{F})$ for any r -graph family $\mathcal{F}$. For $r = 2$, let $f = \min_{F \in \mathcal{F}} \chi(F) - 1$. Since any complete f -partite graph is $\mathcal{F}$ -free, we have $\gamma(\mathcal{F}) \geq 1 - 1/f = \pi(\mathcal{F})$. Hence $\gamma(\mathcal{F}) = \pi(\mathcal{F})$ for every graph family $\mathcal{F}$.

The situation for hypergraphs is far more complicated. There exists an r -graph F for which $\pi(F)$ and $\gamma(F)$ differ almost by 1. For example, fix $r = 3$ and $k \geq 3$, and let F be the 3-graph obtained from a complete graph on k vertices (called *core vertices*) by enlarging each edge with a new (distinct) vertex. It is easy to see that $\text{co-ex}(n, F) \leq \binom{k}{2} + k - 2$ and thus $\gamma(F) = 0$. Indeed, a copy of F can be found in any 3-graph G with $\mathcal{C}(G) \geq \binom{k}{2} + k - 2$ by arbitrarily picking a set S of k vertices and then choosing distinct common neighbors outside S for each pair of S . On the other hand, let G_n be the 3-graph whose vertices are equally partitioned into $k - 1$ sets and whose edges are the triples intersecting each partition set in at most one vertex. Clearly G_n does not contain F , since two of the core vertices of F must lie in the same partition set and they have no neighbor. Since $e(G_n) > \frac{(k-2)(k-3)}{(k-1)^2} \binom{n}{3}$, we conclude that $\pi(F) \geq (k-2)(k-3)/(k-1)^2 \rightarrow 1$ as $k \rightarrow \infty$.

In the opposite direction, for every even $r \geq 4$, there is an r -graph whose π and γ values are the same. Let $T^{(2k)}$ be the $2k$ -graph obtained by letting P_1, P_2, P_3 be pairwise disjoint sets of size k and taking as edges the three sets $P_i \cup P_j$ with $i \neq j$. Keevash and Sudakov [16] determined $\text{ex}(n, T^{(2k)})$ for large n , which in particular, implies that $\pi(T^{(2k)}) = 1/2$. Since the

n -vertex $T^{(2k)}$ -free $2k$ -graph given in [16] has minimum co-degree $n/2 - o(n)$, we conclude that $\gamma(T^{(2k)}) = 1/2 = \pi(T^{(2k)})$.

There are a few 3-graphs whose γ values are known or have been conjectured. One of the trivial examples is $F = \{\{123\}, \{124\}, \{345\}\}$. It is easy to see that $\gamma(F) = 0$ (in contrast, Frankl and Füredi [12] showed that $\pi(F) = 2/9$). Indeed, in any r -graph G_n with $\mathcal{C}(G_n) \geq 3$ we may find a copy of F as follows. First pick any pair of vertices v_1, v_2 , next pick any two vertices v_3, v_4 in their neighborhood $N(v_1, v_2)$, and finally pick a vertex v_5 in $N(v_3, v_4)$ (these choices are possible because both $d(v_1, v_2)$ and $d(v_3, v_4)$ are at least three). The only known nontrivial example is the Fano plane $\mathbf{F}$. The first author recently [17] proved that $\gamma(\mathbf{F}) = 1/2$, in contrast to the result of de Caen and Füredi [4] that $\pi(\mathbf{F}) = 3/4$. Finally, let K_4^3 denote the complete 3-graph on 4 vertices. It was conjectured in [5] that $\gamma(K_4^3) = 1/2$ while the famous Turán Conjecture [23] says that $\pi(K_4^3) = 5/9$.

In fact, for all known 3-graph families $\{G_n\}$ that provide lower bounds for π , there exists $\varepsilon > 0$, such that $e(G_n)/\binom{n}{3} > \mathcal{C}(G_n)/n + \varepsilon$ for sufficiently large n . Hence it would be surprising if $0 < \gamma(F) = \pi(F)$ for any 3-graph F . In any case, it is well-known that determining $\pi(\mathcal{F})$ is very difficult for most families $\mathcal{F}$ (see, *e.g.*, [13] for a survey). Similarly, determining $\gamma(\mathcal{F})$ when $r \geq 3$ does not seem to be much easier.

One of the most important results in extremal hypergraph theory is the so called supersaturation phenomenon, discovered by Brown, Erdős and Simonovits [9]. An indication of its usefulness is that it yields a proof of **ESS** which we sketch below (another way to prove **ESS** is to apply the Regularity Lemma of Szemerédi [22]). Recall that Turán's theorem [23] implies that $\pi(K_k) = 1 - \frac{1}{k-1}$, where K_k is a complete graph on k vertices. Supersaturation implies that for all $\varepsilon > 0$, there exist $\delta = \delta(\varepsilon) > 0$ and n_0 such that if $n > n_0$ and $e(G_n) > (1 - \frac{1}{k-1} + \varepsilon)\binom{n}{2}$, then G_n contains $\delta\binom{n}{k}$ copies of K_k . Another result of Erdős [6] implies that G_n therefore contains a large complete k -partite graph, thus contains all (fixed) graphs with chromatic number at most k ; this is the essential content of **ESS**. As Proposition 1.4 below shows, the supersaturation phenomenon also holds for γ . The sum of sets represents their disjoint union.

Definition 1.3. Let ℓ, n be positive integers and let F be an r -graph on $[h]$. The blow-up $F(\ell)$ is the r -graph (V, E) with $V = V_1 + V_2 + \dots + V_h$, every $|V_i| = \ell$ and $E = \{\{v_{i_1}, v_{i_2}, \dots, v_{i_r}\} : v_{i_j} \in V_{i_j}, \{i_1, i_2, \dots, i_r\} \in E(F)\}$.

For example, blowing up one r -set creates a complete r -partite r -graph $K_r^r(\ell)$.

Proposition 1.4. (Supersaturation) Let F be an r -graph on f vertices. For any $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon) > 0$ and N such that every r -graph G_n with $n > N$ and $c(G_n) > \gamma(F) + \varepsilon$ contains $\delta\binom{n}{f}$ copies of F . Consequently, for every positive integer ℓ , $\gamma(F) = \gamma(F(\ell))$.

For each $r \geq 2$, let

$$\Pi_r = \{\pi(\mathcal{F}) : \mathcal{F} \text{ is a family of } r\text{-graphs}\}.$$

Then **ESS** implies that $\Pi_2 = \{0, \frac{1}{2}, \frac{2}{3}, \dots, \frac{k-1}{k}, \dots\}$. The well-ordered property of Π_2 leads one to the following definition [3, 11, 20] (although there are several equivalent formulations): a real

number $0 \leq a < 1$ is called a *jump* for r if there exists $\delta > 0$, such that no family of r -graphs $\mathcal{F}$ satisfies $\pi(\mathcal{F}) \in (a, a + \delta)$. The set Π_2 shows that every real number in $[0, 1)$ is a jump for $r = 2$, and Erdős conjectured [7] that this is also the case for $r \geq 3$. By supersaturation we have $\pi(K_r^r(\ell)) = 0$. This, together with $\lim_{\ell \rightarrow \infty} e(K_r^r(\ell))/\binom{\ell}{r} = r!/r^r$ implies that no $\mathcal{F}$ satisfies $\pi(\mathcal{F}) \in (0, r!/r^r)$. Thus every $\alpha \in [0, r!/r^r)$ is a jump for $r \geq 3$. A striking result of Frankl and Rödl [11] showed that $1 - 1/\ell^{r-1}$ is not a jump for $r \geq 3$ and $\ell > 2r$, thus disproving Erdős' conjecture. Later Rödl and Sidorenko [20] also answered Erdős' question negatively for multigraphs with edge-multiplicity at least 4.

We consider the same question for the new parameter γ . For $r \geq 2$, let

$$\Gamma_r = \{\gamma(\mathcal{F}) : \mathcal{F} \text{ is a family of } r\text{-graphs}\}.$$

We know that $\Gamma_r \subseteq [0, 1)$ because $\gamma(\mathcal{F}) < 1$ for every family $\mathcal{F}$ of finite r -graphs. Since $\gamma(\mathcal{F}) = \pi(\mathcal{F})$ for all graph families $\mathcal{F}$, we have $\Gamma_2 = \{0, \frac{1}{2}, \frac{2}{3}, \dots, \frac{k-1}{k}, \dots\}$. However, Γ_r behaves differently for $r \geq 3$. One of our main results, Theorem 1.6, implies that Γ_r is dense in $[0, 1)$ for each $r \geq 3$.

Definition 1.5. Fix $r \geq 2$. A real number $0 \leq \alpha < 1$ is called a *jump* for r if there exists $\delta = \delta(\alpha) > 0$, such that every (infinite or finite) family of r -graphs $\mathcal{F}$ satisfies $\gamma(\mathcal{F}) \notin (\alpha, \alpha + \delta)$.

Theorem 1.6. For each $r \geq 3$, no $\alpha \in [0, 1)$ is a jump for r .

Theorem 1.6 implies that Γ_r is dense in $[0, 1)$, however, we believe that more should be true, and that $\Gamma_r = [0, 1)$ for each $r \geq 3$. The missing step for the following conjecture is a compactness property for γ . Note that, in particular, Conjecture 1.7 clearly implies that $\Gamma_r = [0, 1)$.

Conjecture 1.7. Fix $r \geq 3$. For every $0 \leq \alpha < 1$ there exists an infinite family $\mathcal{F}$ of r -graphs such that $\gamma(\mathcal{F}) = \alpha$ and all finite families $\mathcal{F}' \subset \mathcal{F}$ satisfying $\gamma(\mathcal{F}') > \alpha$.

A family $\mathcal{F}$ of r -graphs is called *non-principal* [1, 18] if its Turán density is strictly less than the density of each member. When $r = 2$, **ESS** implies that no family is non-principal because $\pi(\mathcal{F}) = \min_{F \in \mathcal{F}} 1 - \frac{1}{\chi(F)-1} = \min_{F \in \mathcal{F}} \pi(F)$. Motivated by exploring the difference between graphs and hypergraphs, the first author and Rödl [19] conjectured that non-principal families exists for $r \geq 3$. Balogh [1] proved this conjecture by constructing a non-principal 3-graph family with finitely many members. The first author and Pikhurko [18] generalized Balogh's construction to obtain a non-principal family of finite size for all $r \geq 3$, and used new ideas to construct one of size two for $r = 3$ and every even $r \geq 4$. One might suspect that a similar result holds for γ , and our next theorem shows this to be the case.

Theorem 1.8. For every $r \geq 3$, there exists a finite family $\mathcal{F}$ of r -graphs such that $0 < \gamma(\mathcal{F}) < \min_{F \in \mathcal{F}} \gamma(F)$.

The rest of the paper is organized as follows. We prove Propositions 1.2 and 1.4 in Section 2, Theorem 1.6 in Section 3 and Theorem 1.8 in Section 4. In the last section we give some concluding remarks and open problems.

2. Supersaturation

Our goal in this section is to prove Propositions 1.2 and 1.4. Our main tool (Lemma 2.1 below) is a useful technical result used in this section as well as later.

Let $a, \lambda > 0$ with $a + \lambda < 1$. Suppose that $S \subseteq [n]$ and $|S| \geq (a + \lambda)n$. Then a result on hypergeometric distribution (see, e.g., [14] page 29) says that

$$\left| \left\{ M \in \binom{[n]}{m} : |M \cap S| \leq am \right\} \right| \leq \binom{n}{m} e^{-\frac{\lambda^2 m}{3(a+\lambda)}} \leq \binom{n}{m} e^{-\lambda^2 m/3}. \quad (1)$$

For a hypergraph H and a subset $S \subset V(H)$, we denote by $H[S]$ the subhypergraph of H induced by the set S . For positive integers $r < n$, let $[n] = \{1, \dots, n\}$ and $\binom{[n]}{r}$ be the family of all subsets of $[n]$ of size r .

Lemma 2.1. *Fix $r \geq 2$. Given $\varepsilon, \alpha > 0$ with $\alpha + \varepsilon < 1$, let $M(\varepsilon)$ be the smallest integer such that every $m > M(\varepsilon)$ satisfies $m \geq \frac{2(r-1)}{\varepsilon}$ and $\binom{m}{r-1} e^{-\varepsilon^2(m-r+1)/12} \leq \frac{1}{2}$. If $n \geq m \geq M(\varepsilon)$ and G is an r -graph on $[n]$ with $c(G) \geq \alpha + \varepsilon$, then the number of m -sets S satisfying $c(G[S]) > \alpha$ is at least $\frac{1}{2} \binom{n}{m}$. In particular, every r -graph H_n ($n \geq m$) contains a subhypergraph H'_m with $c(H'_m) > c(H_n) - \varepsilon$.*

Proof. Given an $(r-1)$ -set T of $[n]$, we call an m -set S of $[n]$ *bad for T* if $T \subset S$ and $|N(T) \cap S| \leq \alpha m$. We call an m -set S *bad* if it is bad for some T . Let Φ denote the number of bad m -sets, and let Φ_T be the number of m -sets that are bad for T . We need to show that $\Phi \leq \frac{1}{2} \binom{n}{m}$. Clearly

$$\Phi \leq \sum_{T \in \binom{[n]}{r-1}} \Phi_T = \sum_{T \in \binom{[n]}{r-1}} \left| \left\{ S' \in \binom{[n] \setminus T}{m - (r-1)} : |N(T) \cap S'| \leq \alpha m \right\} \right|.$$

Now $\alpha + \varepsilon < 1$ and $m \geq \frac{2(r-1)}{\varepsilon}$ imply that the summand above is upper bounded by

$$\left| \left\{ S' \in \binom{[n] \setminus T}{m - r + 1} : |N(T) \cap S'| \leq \left(\alpha + \frac{\varepsilon}{2} \right) (m - r + 1) \right\} \right|.$$

Applying (1) with $a = \alpha + \varepsilon/2$ and $\lambda = \varepsilon/2$ yields

$$\Phi_T \leq \binom{n - r + 1}{m - r + 1} e^{-(\varepsilon/2)^2(m-r+1)/3}.$$

Finally, we apply the hypothesis $\binom{m}{r-1} e^{-\varepsilon^2(m-r+1)/12} \leq \frac{1}{2}$ to obtain

$$\Phi \leq \binom{n}{r-1} \binom{n - r + 1}{m - r + 1} e^{-(\varepsilon/2)^2(m-r+1)/3} = \binom{n}{m} \binom{m}{r-1} e^{-\varepsilon^2(m-r+1)/12} \leq \frac{1}{2} \binom{n}{m}. \quad \square$$

Call a hypergraph *nontrivial* if it contains at least one edge, and a family of hypergraphs nontrivial if it contains at least one nontrivial member.

Corollary 2.2. *For any $0 < \varepsilon < 1$, define $M(\varepsilon)$ as in Lemma 2.1. Then for every $n > m \geq M(\varepsilon)$ and every non-trivial family $\mathcal{F}$ of r -graphs,*

$$\frac{\text{co-ex}(n, \mathcal{F})}{n} - \frac{\text{co-ex}(m, \mathcal{F})}{m} < \varepsilon.$$

Proof. Since $\mathcal{F}$ is nontrivial, there is an $\mathcal{F}$ -free r -graph H_n with $\mathcal{C}(H_n) = \text{co-ex}(n, \mathcal{F})$. By Lemma 2.1, H_n contains a subhypergraph H'_m with $c(H'_m) > c(H_n) - \varepsilon$. Since H'_m is $\mathcal{F}$ -free, $c(H'_m) \leq \text{co-ex}(m, \mathcal{F})/m$ and the desired inequality follows. $\square$

Proof of Proposition 1.2. Let $a_n = \text{co-ex}(n, \mathcal{F})/n$. Corollary 2.2 says that for every $n > m \geq M(\varepsilon)$, we have $a_n - a_m < \varepsilon$. Since $a_n \geq 0$ for every n , it is easy to see that $\lim_{n \rightarrow \infty} a_n$ exists and equals to $\liminf_{n \rightarrow \infty} a_n$. $\square$

Proof of Proposition 1.4. The proof follows the arguments of Erdős and Simonovits for π , with a suitable application of Lemma 2.1. We sketch the main steps below.

Let $\alpha = \gamma(F)$ and $f = |V(F)|$. For each positive n , let G_n be an r -graph with $\mathcal{C}(G_n) > (\alpha + \varepsilon)n$. By Lemma 2.1, there exists an integer m , such that for $n \geq m$ at least $\frac{1}{2} \binom{n}{m}$ induced subgraphs of G_n on m vertices have minimum co-degree at least $(\alpha + \varepsilon/2)m$. Since $\gamma(F) = \alpha$ and m is sufficiently large, each of these subgraphs contains a copy of F . Consider an f -uniform graph G' on $V(G_n)$ whose edge are f -sets S in which $G[S] \supseteq F$. Then

$$e(G') \geq \frac{1}{2} \frac{\binom{n}{m}}{\binom{n-f}{m-f}} = \frac{1}{2} \frac{\binom{n}{m}}{\binom{m}{f}} \binom{n}{f} = \delta \binom{n}{f}.$$

A result of Erdős [6] implies that for each L , there is a sufficiently large n such that $\mathcal{K} = K_f^f(L) \subseteq G'$. Furthermore, each edge $e = (v_1, v_2, \dots, v_f) \in E(\mathcal{K})$ corresponds to an embedding of F and the mapping of $[f] = V(F)$ to $v_1, v_2, \dots, v_f$ is regarded as a permutation ρ_e of $[f]$. A result in Ramsey theory says that if L is large enough, then we can always find $\mathcal{K}' = K_f^f(\ell) \subseteq \mathcal{K}$ such that all ℓ^f edges in $\mathcal{K}'$ follow the same permutation. This implies that for n sufficiently large the induced subgraph $G[V(\mathcal{K}')] contains a copy of $F(\ell)$. Therefore $\gamma(F) = \gamma(F(\ell))$. $\square$$

3. Jumps

We begin by giving three equivalent definitions for jumps.

Proposition 3.1. *Fix $r \geq 2$. Let $0 \leq \alpha < 1$, $0 < \delta \leq 1 - \alpha$. The following statements are equivalent.*

S1: *Every family of r -graphs $\mathcal{F}$ satisfies $\gamma(\mathcal{F}) \notin (\alpha, \alpha + \delta)$.*

S2: *Every finite family of r -graphs $\mathcal{F}$ satisfies $\gamma(\mathcal{F}) \notin (\alpha, \alpha + \delta)$.*

S3: For every $\varepsilon > 0$ and every $M \geq r - 1$, there exists an integer N such that, for every r -graph G_n with $n > N$ and $\mathcal{C}(G_n) \geq (\alpha + \varepsilon)n$, we can find a subhypergraph $G'_m \subseteq G_n$ with $\mathcal{C}(G'_m) \geq (\alpha + \delta - \varepsilon)m$ for some $m > M$ (Note that the order of quantifiers above is $\forall \varepsilon, M \exists N \forall n > N \exists m > M$).

Remark. A slightly stronger statement than **S3** was stated in the abstract of [11]. There the factor $\alpha + \delta - \varepsilon$ was replaced by $\alpha + \delta$, and the quantification $\forall M \geq r - 1, \exists m > M, G'_m$ was replaced by $\forall M \geq r - 1, \exists G'_M$. The stronger statement was valid in that context because of the monotonicity of $\text{ex}(n, \mathcal{F})/\binom{n}{r}$. As mentioned in the introduction, we could not prove that $\text{co-ex}(n, \mathcal{F})/n$ is monotone, hence we have the different but essentially equivalent statement **S3**.

In Section 3.1 we prove Proposition 3.1. The proof of Theorem 1.6 is then divided into two cases: $\alpha = 0$ (Section 3.2) and $0 < \alpha < 1$ (Section 3.3).

3.1. Proof of Proposition 3.1

We need the following so-called *Continuity property* (which holds for π as shown in [3, 20]).

Lemma 3.2. Let $\mathcal{F}$ be a family of r -graphs. For every $\varepsilon > 0$, there exists a finite family $\mathcal{F}' \subseteq \mathcal{F}$ with $\gamma(\mathcal{F}) \leq \gamma(\mathcal{F}') \leq \gamma(\mathcal{F}) + \varepsilon$.

Proof. Trivially $\gamma(\mathcal{F}) \leq \gamma(\mathcal{F}')$ for any $\mathcal{F}' \subseteq \mathcal{F}$, so we only need to show that $\gamma(\mathcal{F}') \leq \gamma(\mathcal{F}) + \varepsilon$ for some finite family $\mathcal{F}' \subseteq \mathcal{F}$. Set $\gamma = \gamma(\mathcal{F})$ and choose $m = m(\varepsilon)$ such that

1. $\frac{\text{co-ex}(m, \mathcal{F})}{m} < \gamma + \frac{\varepsilon}{2}$ and
2. $m > M(\frac{\varepsilon}{2})$, where $M(\varepsilon)$ is defined as in Corollary 2.2.

Let $\mathcal{F}'$ be the set of members of $\mathcal{F}$ on at most m vertices. Then $\text{co-ex}(m, \mathcal{F}) = \text{co-ex}(m, \mathcal{F}')$. Now we apply Corollary 2.2 to derive that for every $n > m$,

$$\frac{\text{co-ex}(n, \mathcal{F}')}{n} < \frac{\text{co-ex}(m, \mathcal{F}')}{m} + \frac{\varepsilon}{2} = \frac{\text{co-ex}(m, \mathcal{F})}{m} + \frac{\varepsilon}{2} < \gamma + \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \gamma + \varepsilon.$$

Therefore $\gamma(\mathcal{F}') = \lim_{n \rightarrow \infty} \frac{\text{co-ex}(n, \mathcal{F}')}{n} \leq \gamma + \varepsilon$. □

Proof of Proposition 3.1.

Trivially **S1** $\Rightarrow$ **S2**. We will show **S2** $\Rightarrow$ **S1**, **S3** $\Rightarrow$ **S1** and **S1** $\Rightarrow$ **S3**.

S2 $\Rightarrow$ **S1**. Assume that there exists $\delta > 0$, such that no finite family of r -graphs $\mathcal{F}$ satisfies $\gamma(\mathcal{F}) \in (\alpha, \alpha + \delta)$. Suppose that **S1** does not hold, i.e., there exists a family of r -graphs $\mathcal{F}$ satisfying $\gamma(\mathcal{F}) \in (\alpha, \alpha + \delta)$. Let $0 < \varepsilon < \alpha + \delta - \gamma(\mathcal{F})$. We apply Lemma 3.2 to obtain a finite family $\mathcal{F}' \subseteq \mathcal{F}$ with $\gamma(\mathcal{F}') \leq \gamma(\mathcal{F}) + \varepsilon < \alpha + \delta$, a contradiction.

S3 $\Rightarrow$ **S1**. Suppose that **S3** holds. We will show that no family of r -graphs $\mathcal{F}$ satisfies $\gamma(\mathcal{F}) \in (\alpha, \alpha + \delta)$. Suppose instead, that there exist a family $\mathcal{F}$ satisfying $\gamma(\mathcal{F}) = \alpha + b$ for some $0 < b < \delta$. Set $\varepsilon_0 = \min\{\frac{b}{2}, \frac{\delta-b}{2}\}$. Then there exists $N_1 = N_1(\varepsilon)$ so that the following two statements hold:

D1: For every $n > N_1$, there exists an $\mathcal{F}$ -free hypergraph H_n with $c(H_n) \geq \alpha + \varepsilon_0$.

D2: Every hypergraph G_m with $m > N_1$ and $c(G_m) \geq \alpha + b + \varepsilon_0$ contain a member of $\mathcal{F}$.

By **S3** (with $\varepsilon = \varepsilon_0$, $M = N_1$), we may find an n such that the $\mathcal{F}$ -free hypergraph H_n in D1 contains an m -vertex subhypergraph G'_m with $c(G'_m) \geq \alpha + \delta - \varepsilon_0 > \alpha + b + \varepsilon_0$ for some $m > N_1$. This contradicts D2 because $G_m \subset H_n$ is $\mathcal{F}$ -free.

S1 $\Rightarrow$ **S3**. Suppose that **S1** holds but **S3** does not. If **S3** is false for $\varepsilon > 0$, then it is also false for $\varepsilon' < \varepsilon$. Consequently, we may assume that there exist $0 < \varepsilon < \delta$, $M \geq r - 1$, and a sequence of hypergraphs H_{n_i} ($n_i \rightarrow \infty$ as $i \rightarrow \infty$) such that

P1: $c(H_{n_i}) \geq \alpha + \varepsilon$,

P2: $c(H'_m) < \alpha + \delta - \varepsilon$ for every m -vertex hypergraph $H' \subseteq H_{n_i}$ with $m > M$.

Let $\mathcal{F} = \{G : G \not\subseteq H_{n_i} \text{ for any } i\}$. Note that $\mathcal{F}$ is nonempty because P2 implies that $K_m \not\subseteq H_{n_i}$ for every i , thus $K_m \in \mathcal{F}$ (for every $m > M$). Then $\text{co-ex}(n_i, \mathcal{F}) \geq (\alpha + \varepsilon)n_i$ for every i because H_{n_i} is $\mathcal{F}$ -free. Thus $\gamma(\mathcal{F}) \geq \alpha + \varepsilon$. From **S1**, we know that $\gamma(\mathcal{F}) \geq \alpha + \delta$. Hence for every natural number $m > M$, there exists an $\mathcal{F}$ -free hypergraph G_m with $c(G_m) \geq \alpha + \delta - \varepsilon$. Since G_m is $\mathcal{F}$ -free, there exists some n_i such that $G_m \subseteq H_{n_i}$ (otherwise G_m itself is a member of $\mathcal{F}$). But this clearly contradicts P2. $\square$

3.2. Proof of Theorem 1.6 for $\alpha = 0$

Using Statement **S3** in Proposition 3.1, the following shows that 0 is not a jump for r . For every $\delta > 0$, there exist $\varepsilon_0 > 0$ and $M_0 > 0$, such that for every $\ell \geq r - 1$, there exist $n > \ell$ and an r -graph G_n satisfying

(1) $\mathcal{C}(G_n) \geq \varepsilon_0 n$,

(2) $\mathcal{C}(G'_m) < (\delta - \varepsilon_0)m$ for every $G'_m \subset G_n$ with $m > M_0$.

Remark. Note again that the order of quantifiers in the theorem is

$$\forall \delta \quad \exists \varepsilon_0, M_0 \quad \forall \ell \quad \exists n > \ell, G_n \quad \forall G'_m, m > M_0$$

The following special r -graph is important to our proof.

Definition 3.3. $B(\ell, t, r)$ is the r -graph (V, E) with $V = V_1 + V_2 + \cdots + V_t$, each $|V_i| = \ell$ and E comprises all r -sets S with $|S \cap V_i| \leq r - 2$ for all i .

Fix $\delta \in (0, 1)$. Let

$$\varepsilon_0 = \frac{\delta}{3}, \quad M_0 = \frac{3(r-1)}{\delta}, \quad \text{and} \quad t = \left\lfloor \frac{1}{\varepsilon_0} \right\rfloor. \quad (2)$$

Therefore $\varepsilon_0 < 1/3$ and $t \geq 3$.

For every $\ell \geq r-1$, set $n = t\ell$. Starting from the r -graph $B(\ell, t, r)$, we add to the edge set the r -sets with $r-1$ vertices in V_i and one vertex in V_{i+1} (throughout this subsection, all indices are mod t). Denote the resulting r -graph by G_n . It is easy to see that $\mathcal{C}(G_n) \geq \varepsilon_0 n$. In fact, give an $(r-1)$ -set $R \subset V(G_n)$, let $R_i = R \cap V_i$. If $|\{i : R_i \neq \emptyset\}| = 1$, i.e., $R \subset V_i$ for some i , then $N(R) = V_{i+1}$ and $d(R) = \ell \geq \varepsilon_0 n$. Otherwise, the definition of $B(\ell, t, r)$ implies that $d(R) \geq (t-2)\ell \geq \ell \geq \varepsilon_0 t\ell = \varepsilon_0 n$ when $|\{i : R_i \neq \emptyset\}| = 2$, and $d(R) \geq n - (r-1) > \varepsilon_0 n$ when $|\{i : R_i \neq \emptyset\}| \geq 3$.

To complete the proof, we show that

$$\text{For every } m\text{-set } S \text{ with } m > M_0, G' = G_n[S] \text{ satisfies } \mathcal{C}(G') < (\delta - \varepsilon_0)m. \quad (\star)$$

Suppose that $(\star)$ does not hold. Then $\mathcal{C}(G') = \mathcal{C}(G_n[S]) \geq (\delta - \varepsilon_0)m = 2\varepsilon_0 m$ for some m -set S . Let $S_i = S \cap V_i$ for all i . Since $m > M_0 \geq (r-1)/\varepsilon_0 \geq (r-1)t$, by the pigeonhole principle, there is an i_0 and an $(r-1)$ -set $R_0 \subset S_{i_0}$. Because $d_{G'}(R_0) \geq 2\varepsilon_0 m$ and $N(R_0) \subset V_{i_0+1}$, we have $|S_{i_0+1}| \geq 2\varepsilon_0 m$. Since $2\varepsilon_0 m > r-1$, we may repeat this argument to $i_0 + 1$ and conclude $|S_i| \geq 2\varepsilon_0 m$ for all i . But this yields the contradiction $m \geq t 2\varepsilon_0 m \geq (1/\varepsilon_0 - 1) 2\varepsilon_0 m > m$ (using the fact $\varepsilon_0 < 1/3$).

□

3.3. Proof of Theorem 1.6 for $0 < \alpha < 1$

Since the set of rational numbers is dense in the reals, it suffices to show that $\alpha = a/b$ is not a jump for every two positive integers $a < b$. As in the $\alpha = 0$ case, we will show that the negation of **S3** holds for every $\delta > 0$.

Given $\delta > 0$, let ε_0, M_0, t be as in (2). Set

$$\varepsilon = \frac{\varepsilon_0}{b} \quad \text{and} \quad M = \max \left\{ \frac{rb^2}{(\delta - \varepsilon_0)a}, \frac{M_0 + rb}{\delta - \varepsilon_0}, M_0 b \right\}.$$

For every integer $\ell \geq r-1$, set $n = t\ell b$. Let $\mathcal{D}$ be the directed graph on $[b]$ with $E(\mathcal{D}) = \{(i, j) : j = i + 1, \dots, i + a \text{ mod } b\}$. Let $G_{t\ell}$ be the r -graph constructed in Section 3.2. Let $H_n = (V, E)$ be the r -graph obtained from $B(t\ell, b, r)$ by adding

- edges within each V_i so that $H[V_i] \cong G_{t\ell}$, and
- all edges with $r-1$ vertices in V_i and one vertex in V_j whenever $(i, j) \in E(\mathcal{D})$.

We claim that $\mathcal{C}(H) \geq (a/b + \varepsilon)n$. To see this, pick an $(r-1)$ -set $R \subset V$. If $R \subset V_i$ for some i , then $R \cup \{v\} \in E$ for every $v \in \cup_{j=1}^a V_{i+j \pmod{b}}$, and $d_{H[V_i]}(R) \geq \varepsilon_0 t\ell$ because $H[V_i] \cong G_{t\ell}$. Thus

$$d_{H_n}(R) \geq a|V_i| + \varepsilon_0 t\ell = (a + \varepsilon_0)t\ell = (a/b + \varepsilon)n.$$

Next suppose that $\max_i |R \cap V_i| < r-1$. We consider three cases: $a = b-1$, $a = b-2$ and

$a \leq b - 3$. If $a = b - 1$, then the edges of $H[V_i]$ together with the edges of $B(t\ell, b, r)$ yield

$$d_{H_n}(R) \geq n - (r - 1) \geq n - \ell = (b - 1)t\ell + t\ell - \ell > (b - 1)t\ell + \varepsilon_0 t\ell = (a/b + \varepsilon)n,$$

where the second inequality holds because $t > \varepsilon_0 t + 1$. Following a similar reasoning, when $a = b - 2$, we have

$$d_{H_n}(R) \geq n - t\ell - (r - 1) \geq n - t\ell - \ell > (b - 2)t\ell + \varepsilon_0 t\ell = (a/b + \varepsilon)n,$$

and when $a \leq b - 3$, the edges of $B(t\ell, b, r)$ yield

$$d_{H_n}(R) \geq n - 2t\ell - (r - 1) \geq n - 2t\ell - \ell > (b - 3)t\ell + \varepsilon_0 t\ell \geq (a/b + \varepsilon)n.$$

Let S be an m -set with $m > M$ and $H' = H[S]$. Our goal is to show that $\mathcal{C}(H') < (a/b + \delta - \varepsilon)m$, i.e., there exists an $(r - 1)$ -set $R \subset S$ such that $d_{H'}(R) < (a/b + \delta - \varepsilon)m$.

Let $S_i = S \cap V_i$. We first claim that there exists i_0 , such that $|S_{i_0}| \geq r - 1$ and

$$\sum_{j=i_0+1}^{i_0+a} |S_j| < \frac{a}{b}m + rb. \quad (3)$$

In fact, if $\sum_{j=i+1}^{i+a} |S_j| > \frac{a}{b}m$ for all $i \in [b]$, then by averaging over all intervals of length a , we obtain $|S| > \lceil (b\frac{a}{b}m)/a \rceil \geq m$, a contradiction. Next, assume that $\sum_{j=i+1}^{i+a} |S_j| \leq \frac{a}{b}m$ but $|S_i| \leq r - 2$ for some i . Let i_0 be the largest index smaller than $i \pmod{b}$ such that $|S_{i_0}| \geq r - 1$ (such i_0 exists because $m > M > (r - 2)b$). It is easy to see that (3) follows.

Let R be an $(r - 1)$ -subset of S_{i_0} . We will show that $d_{H'}(R) < (a/b + \delta - \varepsilon_0)m < (a/b + \delta - \varepsilon)m$. If $|S_{i_0}| > M_0$, then $H[V_{i_0}] \cong G_{t\ell}$ and $(\star)$ implies that $d_{H'[S_{i_0}]}(R) < (\delta - \varepsilon_0)|S_{i_0}|$. Otherwise $d_{H'[S_{i_0}]}(R) \leq |S_{i_0}| \leq M_0$.

If $|S_{i_0}| \geq (1 - \frac{a}{b})m$, then $b \geq a + 1$ and $m \geq bM_0$ yield $|S_{i_0}| \geq \frac{m}{b} > M_0$. Therefore

$$d_{H'}(R) < (\delta - \varepsilon_0)|S_{i_0}| + (m - |S_{i_0}|) < (\delta - \varepsilon_0)m + \frac{a}{b}m.$$

Otherwise, $m - |S_{i_0}| > \frac{a}{b}m > \frac{rb}{\delta - \varepsilon_0}$, since $m > \frac{rb^2}{(\delta - \varepsilon_0)a}$. Consequently

$$(\delta - \varepsilon_0)|S_{i_0}| + rb < (\delta - \varepsilon_0)m. \quad (4)$$

By the structure of $\mathcal{D}$, we know that all the neighbors of R in $H'[S \setminus S_{i_0}]$ are in S_j , for $j = i_0 + 1, \dots, i_0 + a \pmod{b}$. Applying (3), we therefore obtain $d_{H'[S \setminus S_{i_0}]}(R) \leq \sum_{j=i_0+1}^{i_0+a} |S_j| < \frac{a}{b}m + rb$, and hence

$$\begin{aligned} d_{H'}(R) &= d_{H'[S_{i_0}]}(R) + d_{H'[S \setminus S_{i_0}]}(R) \\ &< \max\{M_0, (\delta - \varepsilon_0)|S_{i_0}|\} + \frac{a}{b}m + rb \\ &= \max\{M_0 + rb, (\delta - \varepsilon_0)|S_{i_0}| + rb\} + \frac{a}{b}m \end{aligned}$$

The hypothesis $m > \frac{M_0 + rb}{\delta - \varepsilon_0}$ implies that $M_0 + rb < (\delta - \varepsilon_0)m$, and together with (4), we again derive that $d_{H'}(R) < (\delta - \varepsilon_0)m + \frac{a}{b}m$. $\square$

4. Non-Principality

In this section, we prove Theorem 1.8 by an explicit construction. An r -graph G is called *2-colorable* (or with *chromatic number 2*) if $V(G)$ can be partitioned into two disjoint sets A and B such that no edges fall insider A or B . The main idea in our proof is to find $\gamma_0 < \frac{1}{2}$ and a 2-colorable r -graph F with $\gamma(F) \geq \frac{1}{2}$ such that every 2-colorable r -graph H_n with $c(H_n) \geq \gamma_0$ contains F as a subgraph.

Definition 4.1. $K^r(t, t)$ is the r -graph with vertex set $A + B$, $|A| = |B| = t$, and edge set $\{S \in \binom{V}{r} : |S \cap A| = 1 \text{ or } |S \cap B| = 1\}$.

Proposition 4.2. For $r \geq 3$, there exists a positive integer $\ell = \ell(r)$ such that $\gamma(K^r(\ell, \ell)) \geq \frac{1}{2}$.

Proof. It suffices to show that for every $\varepsilon > 0$, there exists $N > 0$ such that for every $n > N$, there exists a $K^r(\ell, \ell)$ -free r -graph H_n^r with $c(H_n^r) > \frac{1}{2} - \varepsilon$. For $r = 3$, Nagle and Rödl (see [5]) constructed a K_4^3 -free 3-graph H_n^3 for sufficiently large n with $c(H_n^3) > \frac{1}{2} - \varepsilon$. Since $K_4^3 = K^3(2, 2)$, H_n^3 is the desired 3-graph and $\ell(3) = 2$.

Beginning with H_n^3 above, for $r > 3$ and sufficiently large n , we define the r -graph H_n^r with vertex set $[n]$ and $E(H_n^r) = \{D \in \binom{[n]}{r} : D \supset T \text{ for some } T \in E(H_n^3)\}$.

It is easy to see that $c(H_n^r) > \frac{1}{2} - \varepsilon$ for every $r > 3$. Indeed, consider a set $S \in \binom{[n]}{r-1}$. If S contains some $T \in E(H_n^3)$, then $d_{H_n^r}(S) = n - r$. Otherwise we consider any two vertices $a, b \in S$. Since $N_{H_n^3}(a, b) \cap S = \emptyset$ and $d_{H_n^3}(a, b) \geq \mathcal{C}(H_n^3) > \frac{n}{2} - \varepsilon n$, we have $d_{H_n^r}(S) > \frac{n}{2} - \varepsilon n$. In either case $\mathcal{C}(H_n^r) > \frac{n}{2} - \varepsilon n$ (assuming $n > 2r$) and consequently $c(H_n^r) > \frac{1}{2} - \varepsilon$.

We now prove that H_n^r contains no copy of $K^r(\ell, \ell)$ by induction on r . To facilitate the induction, we prove the stronger statement that H_n^r is not only $K^r(\ell, \ell)$ -free, but also K_k^r -free for some $\ell = \ell(r)$ and $k = k(r)$.

The base case $r = 3$ is certainly true with $\ell = 2$ and $k = 4$. For the induction step, suppose that H_n^r contains no copy of K_p^r or $K^r(q, q)$ for some integers p and q . We need to find two integers k and ℓ such that H_n^{r+1} contains no K_k^{r+1} or $K^{r+1}(\ell, \ell)$. Let us first note that the definition of H_n^r implies that for every $S \in \binom{[n]}{r+1}$,

$$S \in E(H_n^{r+1}) \Leftrightarrow \exists T \subset S, T \in E(H_n^r), \quad (5)$$

Let k be the (hypergraph) Ramsey number $R^r(p, r+1)$. For any k -set S of $[n]$, since $H_n^r[S]$ contains no copy of K_p^r , Ramsey's theorem says that there exists an $(r+1)$ -set $T \subset S$ such that $H_n^r[T]$ has no edges. By (5), T is not an edge of H_n^{r+1} . We thus conclude that H_n^{r+1} contains no K_k^r .

The value of ℓ is obtained in a more complicated way, in which (hypergraph) Ramsey numbers are used iteratively. We need two auxiliary sequences $\{m_i\}_0^q, \{\ell_i\}_0^m$:

$$\begin{aligned} m_0 &= q, & m_i &= R^{r-1}(r, m_{i-1}), & i &= 1, \dots, q; \\ \ell_0 &= q, & \ell_i &= R^{r-1}(r, \ell_{i-1}), & i &= 1, \dots, m = m_q, \end{aligned}$$

and $\ell = R^r(p, \ell_m)$.

Let A and B be two disjoint ℓ -sets of $[n]$. We claim that if $H_n^{r+1}[A \cup B] \cong K^{r+1}(\ell, \ell)$, then $H_n^r[A \cup B]$ contains a copy of $K^r(q, q)$, contradicting the induction hypothesis. To prove it, we first find two ℓ_m -sets $A_0 \subseteq A$ and $B' \subseteq B$ such that A_0 and B' are two independent sets in H_n^r . These sets exist because $\ell = R^r(p, \ell_m)$, and neither $H_n^r[A]$ nor $H_n^r[B]$ contains a copy of K_p^r by the induction hypothesis.

Suppose $B' = \{b_1, b_2, \dots, b_{\ell_m}\}$ and consider b_1 first. Since $S \cup \{b_1\} \in E(H_n^{r+1})$ for every r -set S in A_0 , by (5), there exists an $S' \in \binom{S \cup \{b_1\}}{r}$ such that $S' \in E(H_n^r)$. Since S itself is not an edge of H_n^r (A is independent), we have $S' \neq S$, or $|S' \cap S| = r - 1$. Let G_1 be the link graph of H_n^r for b_1 on A_0 , i.e., $V(G_1) = A_0$, $E(G_1) = \{T \subset A_0 : T \cup \{b_1\} \in E(H_n^r)\}$. We thus know that G_1 contains no independent r -sets on A_0 . Since $\ell_m = R^{r-1}(r, \ell_{m-1})$, there exists an ℓ_{m-1} -set $A_1 \subset A_0$ such that $G_1[A_1] \cong K^{r-1}(\ell_{m-1})$. Repeating this process for $b_2, \dots, b_m$, we find subsets $(A_1 \supset) A_2 \supset \dots \supset A_m$ and link graphs $G_2, \dots, G_m$ for $b_2, \dots, b_m$, respectively, such that $G_i[A_i] \cong K^{r-1}(\ell_{m-i})$. Let $B_0 = \{b_i, 1 \leq i \leq m\}$. We thus know that every union of an $(r-1)$ -subset of A_m and a vertex in B_0 is an edge of H_n^r . Now assume $A_m = \{a_1, \dots, a_q\}$ and apply the same arguments to the link graphs of H_n^r for a_i on B_0 . We obtain a q -set $B_q \subset B_0$ such that the union of every $(r-1)$ -subset of B_q and a vertex in A_m is an edge of H_n^r . In other words, $H_n^r[A_m \cup B_q] \cong K^r(q, q)$. $\square$

Proposition 4.3. *Let $r \geq 3$, $\ell \geq r - 1$ and $\rho = \frac{1}{2} \left(1 - \frac{1}{\binom{\ell}{r-1}} + \frac{1}{\binom{\ell}{r-1}^{2^{1/\ell}}} \right) < \frac{1}{2}$. For any $\varepsilon > 0$ there exists N such that every 2-colorable r -graph G_n with $n > N$ and $\mathcal{C}(G_n) \geq (\rho + \varepsilon)n$ contains a copy of $K^r(\ell, \ell)$.*

Proof. Suppose to the contrary, that for arbitrarily large n , there exists a 2-colorable r -graph G_n such that

- $\mathcal{C}(G_n) = c \geq (\rho + \varepsilon)n$, and
- G_n contains no copy of $K^r(\ell, \ell)$.

Since G_n is 2-colorable, we may partition $V(G_n)$ into two sets A and B with $|A| = a \leq b = |B|$ such that no edges of G_n fall inside A or B . Thus for any $X \in \binom{A}{r-1}$, we have $N(X) \subseteq B$ and the same holds for $Y \in \binom{A}{r-1}$. This implies that $c \leq a \leq b \leq n - c$.

Let $X \subset A$ be an ℓ -set. We first estimate $|\bigcap_{X' \subseteq \binom{X}{r-1}} N(X')|$. Since every $X' \subset \binom{A}{r}$ has at most $b - c$ non-neighbors in B , the number of their common neighbors is at least $b - \binom{\ell}{r-1}(b - c)$. Similarly $|\bigcap_{Y' \subseteq \binom{Y}{r-1}} N(Y')| \geq a - \binom{\ell}{r-1}(a - c)$ for every $Y \in \binom{B}{\ell}$ (the inequality $a - \binom{\ell}{r-1}(a - c) \geq b - \binom{\ell}{r-1}(b - c) > 0$ can be seen from our arguments on the function $f(x)$ defined later).

The key to our proof is to estimate Φ , the number of $X \cup Y$ with $X \in \binom{A}{\ell}$ and $Y \in \binom{B}{\ell}$ such that either

- $X' \cup \{y\} \in E(G_n)$ for all $X' \subseteq \binom{X}{r}$ and $y \in Y$ or

- $Y' \cup \{x\} \in E(G_n)$ for all $Y' \subseteq \binom{Y}{r}$ and $x \in X$.

Trivially $\Phi \leq \binom{a}{\ell} \binom{b}{\ell}$. On the other hand, because G_n does not contain $K^r(\ell, \ell)$, we have

$$\begin{aligned} \Phi &\geq \sum_{X \in \binom{A}{\ell}} \left(\left| \bigcap_{X' \subseteq \binom{X}{r-1}} N(X') \right| \right) + \sum_{Y \in \binom{B}{\ell}} \left(\left| \bigcap_{Y' \subseteq \binom{Y}{r-1}} N(Y') \right| \right) \\ &\geq \binom{a}{\ell} \binom{b - \binom{\ell}{r-1}(b-c)}{\ell} + \binom{b}{\ell} \binom{a - \binom{\ell}{r-1}(a-c)}{\ell} \\ &= \binom{a}{\ell} \binom{ta - (tn - (t+1)c)}{\ell} + \binom{b}{\ell} \binom{tb - (tn - (t+1)c)}{\ell}, \end{aligned}$$

where $t = \binom{\ell}{r-1} - 1 \geq 0$ (equality holds if and only if $\ell = r - 1$).

For fixed t, c, n , define the function $f(x) = \binom{x}{\ell} (tx - (tn - (t+1)c))$ for $x \in [c, n - c]$. After rewriting,

$$f(x) = \frac{\prod_{i=0}^{l-1} (x - i) \prod_{i=0}^{l-1} (tx - (tn - (t+1)c) - i)}{(l!)^2}.$$

We claim that the second derivative $f''(x) \geq 0$ for $x \in [c, n - c]$. By differentiation, this claim holds as long as each term in the products in the numerator is nonnegative and has nonnegative derivative. Since n is sufficiently large, $x - l + 1 \geq c - l + 1 > 0$ so each term in the first product is positive. To show the same for each term in the second product, it suffices to show that $tc \geq tn - (t+1)c + (\ell - 1)$. Using $c > \rho n$, a short calculation shows that this is equivalent to $\frac{1}{2} \left(1 - \frac{1}{t+1} + \frac{1}{2^{1/\ell}(t+1)} \right) > \frac{t}{2t+1}$. This holds since

$$\frac{1}{2} \left(1 - \frac{1}{t+1} + \frac{1}{2^{1/\ell}(t+1)} \right) > \frac{1}{2} \left(1 - \frac{1}{t+1} + \frac{1}{2(t+1)} \right) > \frac{1}{2} \left(1 - \frac{1}{2t+1} \right) = \frac{t}{2t+1}.$$

The derivatives of the terms are 1 and t , which are both nonnegative. We therefore conclude that $f''(x) \geq 0$ and consequently $f(x)$ is convex on $[c, n - c]$. Hence

$$\Phi \geq f(a) + f(b) \geq 2f\left(\frac{a+b}{2}\right) = 2f(n/2) = 2 \binom{n/2}{\ell} \binom{(t+1)c - tn/2}{\ell}.$$

On the other hand, since $\ln \binom{x}{l}$ is a concave function, we have $\Phi \leq \binom{a}{l} \binom{b}{l} \leq \left(\binom{n/2}{l} \right)^2$. Putting the lower and upper bounds for Φ together yields

$$2 \binom{n/2}{\ell} \binom{(t+1)c - tn/2}{\ell} \leq \left(\binom{n/2}{\ell} \right)^2,$$

But this implies that as $n \rightarrow \infty$,

$$c \leq \frac{n}{2} \left(1 - \frac{1}{t+1} + \frac{1}{(t+1)2^{1/\ell}} \right) + o(n) = \frac{n}{2} \left(1 - \frac{1}{\binom{\ell}{r-1}} + \frac{1}{\binom{\ell}{r-1}2^{1/\ell}} \right) + o(n) = \rho n + o(n),$$

contradicting the assumption that $c \geq (\rho + \varepsilon)n$. $\square$

Proof of Theorem 1.8. Let $\ell = \ell(r)$ be as in Proposition 4.2 and ρ as in Proposition 4.3. For any $0 < \varepsilon < \frac{1}{2} - \rho$ we will construct a finite family $\mathcal{F}$ of r -graphs such that

$$\frac{1}{4} \leq \gamma(\mathcal{F}) \leq \rho + \varepsilon < \frac{1}{2} \leq \min_{F \in \mathcal{F}} \gamma(F). \quad (6)$$

Let $m = \max\{M(\varepsilon/2), N(\varepsilon/2)\} + 1$, where M is the threshold function in Lemma 2.1 and N is the threshold function in Proposition 4.3. Let $\mathcal{F}_0$ be the family of r -graphs on at most m vertices which are not 2-colorable. We observe that $\min_{F \in \mathcal{F}_0} \gamma(F) \geq \gamma(\mathcal{F}_0) \geq 1/2$. In fact, for any n , the following r -graph G_n is 2-colorable and satisfies $\mathcal{C}(G_n) = \lfloor n/2 \rfloor$: $V(G_n)$ contains two disjoint vertex sets A and B of sizes differing by at most 1, $E(G_n)$ contains all the edges intersecting both A and B .

We now show that (6) holds for $\mathcal{F} = \mathcal{F}_0 \cup \{K^r(\ell, \ell)\}$. Proposition 4.2 says that $\gamma(K^r(\ell, \ell)) \geq 1/2$. Together with $\min_{F \in \mathcal{F}_0} \gamma(F) \geq 1/2$, we conclude that $\min_{F \in \mathcal{F}} \gamma(F) \geq 1/2$. On the other hand, we claim that $\frac{1}{4} \leq \gamma(\mathcal{F}) \leq \rho + \varepsilon$ and thus (6) follows.

To see that $\gamma(\mathcal{F}) \geq \frac{1}{4}$, let H_n^r be the $K^r(\ell, \ell)$ -free r -graph as in the proof of Proposition 4.2. We randomly partition $V(H_n^r)$ into two almost equal parts and remove all the edges within one part. The resulting r -graphs $\tilde{H}_n^r$ is also $K^r(\ell, \ell)$ -free and satisfies $\mathcal{C}(\tilde{H}_n^r) \geq \frac{n}{4} - o(n)$ (using standard Chernoff bounds). Hence $\gamma(\mathcal{F}) \geq \frac{1}{4}$.

To see that $\gamma(\mathcal{F}) \leq \rho + \varepsilon$, let G_n be an r -graph with $n > m$ and $\mathcal{C}(G_n) \geq (\rho + \varepsilon)n$. By Lemma 2.1, G_n has a subgraph G'_m with $\mathcal{C}(G'_m) \geq (\rho + \varepsilon/2)m$. If G'_m is not 2-colorable, then G'_m itself is a member of $\mathcal{F}$. Otherwise, since $m > N(\varepsilon/2)$ and $\mathcal{C}(G'_m) \geq (\rho + \varepsilon/2)m$, Proposition 4.3 guarantees that G'_m contains a copy of $K^r(\ell, \ell)$. Therefore G always contains a member of $\mathcal{F}$ as a subgraph. Consequently $\text{co-ex}(n, \mathcal{F}) \leq (\rho + \varepsilon)n$ for all $n > m$ and thus $\gamma(\mathcal{F}) \leq \rho + \varepsilon$. $\square$

5. Concluding Remarks and open problems

- Theorem 1.6 and Proposition 3.1 together imply that the set $\{\gamma(\mathcal{F}) : \mathcal{F} \text{ is a finite family}\}$ is dense on $[0, 1)$, i.e., for all $0 \leq \alpha < \beta < 1$, there exists a finite family of r -graphs such that $\gamma(\mathcal{F}) \in (\alpha, \beta)$. It would be interesting to describe the set of $\gamma(F)$ for a single r -graph F . For example, does Theorem 1.6 still hold when $\mathcal{F}$ in Definition 1.5 is replaced by a single r -graph F ? This question is also related to the principality: if there exist $0 \leq \alpha < \beta < 1$ such that $\gamma(F) \notin (\alpha, \beta)$ for every r -graph F , then *every* finite family $\mathcal{F}$ with $\gamma(\mathcal{F}) \in (\alpha, \beta)$ (such $\mathcal{F}$ exists by Theorem 1.6 and Proposition 3.1) is non-principal.
- We have mentioned the problem of verifying $\gamma(K_4^3) < \pi(K_4^3)$ in our introduction. Applying Proposition 4.3 with $r = 3$ and $\ell = 2$, we obtain that every 2-colorable 3-graph G_n with $c(G_n) > \frac{1}{2\sqrt{2}}$ (and large n) contains a copy of K_4^3 . Is this constant $\frac{1}{2\sqrt{2}}$ sharp here? From (6) we know it can not be reduced to a number smaller than $1/4$.

- Parallel to the situation for π , it would be interesting to construct two r -graphs F_1, F_2 such that $0 < \gamma(\{F_1, F_2\}) < \min\{\gamma(F_1), \gamma(F_2)\}$ (Sudakov suggested such a construction for even $r \geq 4$ by following the ideas in [18]).

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